

PROJECT-Q Workbook - 11

30-Day Quantum Computing Challenge

DAY - 11

Quantum Entanglement

Learning Objectives

By the end of today's session, you will be able to:

- ✓ Explain what quantum entanglement is in simple terms.
 - ✓ Differentiate between **classical correlation** and **quantum entanglement**.
 - ✓ Understand how **Hadamard** and **CNOT** gates work together to create entanglement.
 - ✓ Interpret the first Bell State.
 - ✓ Create an entangled pair using Qiskit.
 - ✓ Understand why entanglement is a key resource in quantum computing.
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Quick Recap

Before learning entanglement, let's connect everything you've studied so far.

- A **qubit** can exist in a superposition of $|0\rangle$ and $|1\rangle$.
- Multiple qubits form a larger quantum system whose state is represented together.
- The **Hadamard (H) Gate** creates superposition.
- The **CNOT Gate** allows one qubit to control another.

Today, we'll combine these ideas to create a phenomenon that has **no equivalent in classical computing**.

Why Does Entanglement Matter?

Suppose you have two classical coins. Even if you flip them together, each coin still has its own independent state. Knowing the result of one coin doesn't automatically tell you the state of the other. Quantum systems behave differently. When two qubits become **entangled**, they no longer behave like separate objects. Instead, they behave as **one shared quantum system**. This property enables many of the most powerful applications of quantum computing, including quantum communication, teleportation, cryptography, and several quantum algorithms.

What is Quantum Entanglement?

Quantum entanglement is a phenomenon in which two or more qubits become linked so that the complete quantum state belongs to the system as a whole, rather than to each qubit individually.

This means:

- The qubits cannot always be described independently.
- Measuring one qubit provides information about the other.
- The relationship exists because they share the same quantum state.

Key Idea: Entanglement is about **shared quantum states**, not faster communication.

A Common Misconception

Many people say:

"If I measure one qubit, it instantly sends information to the other."

This is **not** what quantum mechanics says.

Instead:

The qubits were already prepared in a **shared quantum state**. Measurement reveals the correlation that already existed it does **not** create or transmit it.

Classical Correlation vs Quantum Entanglement

At first glance, both ideas seem similar because they involve correlated outcomes. However, they are fundamentally different.

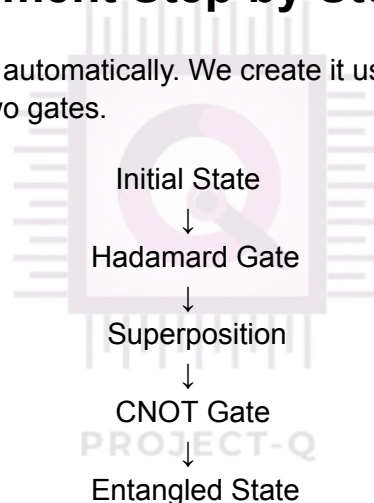
Classical Correlation	Quantum Entanglement
Each object has its own state.	The qubits share one quantum state.
Correlation comes from shared history.	Correlation comes from a shared quantum description.
Can be explained using classical physics.	Requires quantum mechanics.
Measuring one object reveals only its own state.	Measuring one qubit determines the correlated outcome of the pair.

Think of it this way:

- Classical systems are like **two synchronized clocks**.
- Entangled qubits behave like **one quantum system observed from two locations**.

Building Entanglement Step by Step

Entanglement does not appear automatically. We create it using a quantum circuit. The simplest circuit contains only two gates.



Every Bell State begins with this basic idea.

Step 1 - Start with Two Qubits

Initially, $|00\rangle$

Both qubits are in the state $|0\rangle$.

Nothing unusual yet.

Step 2 - Apply the Hadamard Gate

Apply an **H Gate** to the first qubit.

The first qubit enters an equal superposition.

The combined system becomes

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

Notice that the second qubit has not changed.

Only the first qubit is in superposition.

Step 3 - Apply the CNOT Gate

Now apply a CNOT gate.

- Control → First qubit
- Target → Second qubit

The CNOT flips the second qubit only when the control qubit is $|1\rangle$.

The resulting state becomes

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This is called the **Bell State** $|\Phi^+\rangle$.

More importantly,

it is an **entangled state**.

Understanding the Bell State

The Bell State

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

contains only two possible outcomes:

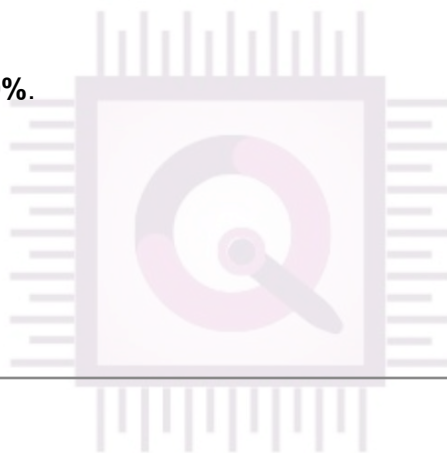
- $|00\rangle$
- $|11\rangle$

Each occurs with probability **50%**.

The outcomes

$|01\rangle$ and $|10\rangle$

never occur in an ideal system.



Visualizing the Measurement

Suppose you run the circuit 1000 times.

A typical result might look like

State	Counts
00	501
11	499
01	0
10	0

Notice that the results are **random**, but they are also **perfectly correlated**.

You cannot predict whether the next measurement will be **00** or **11**.

However, once one qubit is measured, the value of the other is immediately determined by the shared state.

Why Isn't This Just a Superposition?

This is an important question.

A single qubit in superposition behaves like

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Each qubit can still be described individually.

An entangled pair cannot.

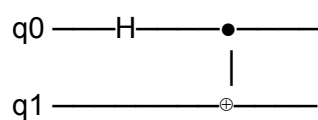
The Bell State

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

describes the **entire system**, not each qubit separately.

This is the defining feature of entanglement.

Circuit Representation



- The Hadamard gate creates superposition.
 - The CNOT gate links the two qubits.
 - Together they produce an entangled pair.
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Qiskit Implementation

Step 1 - Import Qiskit

```
from qiskit import QuantumCircuit
```

Step 2 - Create the Circuit

```
qc = QuantumCircuit(2,2)
```

Step 3 - Create Superposition

```
qc.h(0)
```

Step 4 - Apply the CNOT Gate

```
qc.cx(0,1)
```

Step 5 - Measure Both Qubits

```
qc.measure([0,1],[0,1])
```

Complete Program

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2,2)
qc.h(0)
qc.cx(0,1)
qc.measure([0,1],[0,1])
print(qc.draw())
```



Expected Output

When executed on an ideal simulator, you should observe results similar to:

```
00 : 506
11 : 494
```

The outputs **01** and **10** should not appear.

When running on real quantum hardware, small deviations may occur because physical qubits are affected by noise and decoherence.

Why is Entanglement Important?

Entanglement is not just an interesting property it is a practical resource.

Many quantum technologies rely on it, including:

- Quantum teleportation

- Quantum cryptography
- Superdense coding
- Quantum error correction
- Quantum networking
- Variational quantum algorithms
- Many quantum machine learning techniques

Without entanglement, quantum computers would lose much of their computational advantage.

Real-World Connection

Researchers use entangled photons in secure communication systems.

Because measuring an entangled quantum system changes its state, any attempt to intercept the communication can be detected.

This principle forms the basis of **Quantum Key Distribution (QKD)**, one of the first commercially explored applications of quantum information science.

Common Beginner Mistakes

✗ "Entangled qubits always have opposite values."

✓ Not necessarily.

Different entangled states produce different correlations.

✗ "Entanglement allows faster-than-light communication."

✓ No.

Entanglement creates correlations, but it cannot be used to transmit information instantly.

✗ "Superposition and entanglement are the same."

✓ No.

Superposition describes the state of an individual qubit.

Entanglement describes a relationship between multiple qubits.

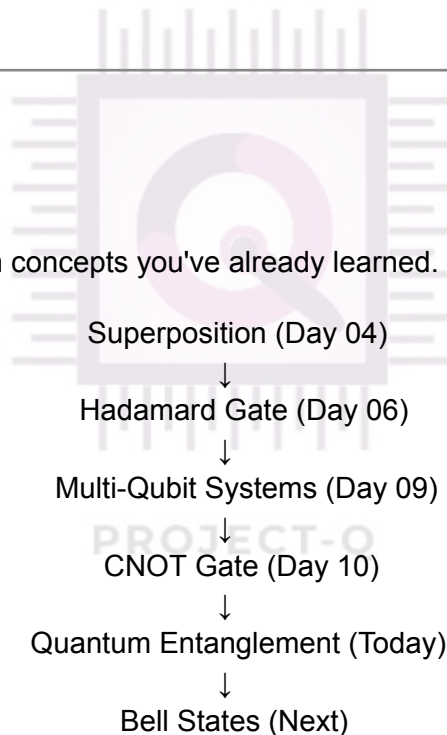
✗ "CNOT alone creates entanglement."

✓ Not always.

If the control qubit is **not** in superposition, a CNOT gate alone does not produce an entangled state. The combination of **Hadamard followed by CNOT** is what creates the Bell state introduced today.

Connections

This chapter builds directly on concepts you've already learned.



Understanding this sequence is more valuable than memorizing definitions because each topic naturally builds on the previous one.

Remember This

- Entanglement is a property of a **combined quantum system**.
 - Entangled qubits cannot always be described independently.
 - The Hadamard gate creates superposition.
 - The CNOT gate transfers that superposition into a shared quantum state.
 - The Bell State ($\frac{|00\rangle + |11\rangle}{\sqrt{2}}$) is the simplest example of an entangled state.
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FORMULA SHEET
DAY 11 – ENTANGLEMENT

1. BASIC QUBIT STATES

State	Description
$ 0\rangle$	Qubit in state 0
$ 1\rangle$	Qubit in state 1
$\alpha 0\rangle + \beta 1\rangle$	General single-qubit state, $ \alpha ^2 + \beta ^2 = 1$

2. TWO-QUBIT COMPUTATIONAL BASIS

The basis states for two qubits are:

$|00\rangle \quad |01\rangle \quad |10\rangle \quad |11\rangle$

GENERAL TWO-QUBIT STATE

$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$
with $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$

3. HADAMARD GATE (H)

Creates superposition.

$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
 $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
 $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

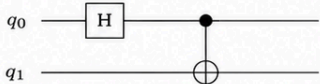
4. CNOT GATE

Flips the target qubit if the control qubit is $|1\rangle$.

Control (c)  Target (t) $\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Basis order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

5. ENTANGLEMENT CREATION (BELL STATE PREPARATION)

Circuit: Apply H to the first qubit, then CNOT (control = q_0 , target = q_1).



Step 1 (Initial State): $|\psi_0\rangle = |00\rangle$
Step 2 (After H on q_0): $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
Step 3 (After CNOT): $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

6. BELL STATE ($|\Phi^+\rangle$)

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Measurement Probabilities:

Outcome	$ 00\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$
Probability	1/2	0	0	1/2



Outcomes $|01\rangle$ and $|10\rangle$ do not occur (in ideal case).
Results $|00\rangle$ and $|11\rangle$ are perfectly correlated.

7. GENERAL BELL STATES

The four maximally entangled Bell states:

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ (Same)
 $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$ (Same, with phase flip)
 $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ (Opposite)
 $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ (Opposite, with phase flip)

8. KEY POINTS (SUMMARY FORMULAS)

- Normalization (2 qubits): $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$
- Superposition after H on first qubit: $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
- Entangled Bell state ($|\Phi^+\rangle$): $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- Measurement probabilities for $|\Phi^+\rangle$: $P(|00\rangle) = \frac{1}{2}$, $P(|11\rangle) = \frac{1}{2}$
- No faster-than-light communication.
- Entanglement is a property of the combined system.



REMEMBER: Entanglement means the qubits share one quantum state.
Measuring one qubit determines the correlated outcome of the other.

End-of-Day Summary

Today, you learned:

- What quantum entanglement is and why it is unique.
- The difference between classical correlation and quantum entanglement.

- How the Hadamard and CNOT gates work together to produce an entangled state.
 - How to prepare and measure the first Bell state using Qiskit.
 - Why entanglement is central to modern quantum computing and communication.
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Top Verified Learning Resources

1. Real Python - Quantum Computing Basics

<https://realpython.com/quantum-computing-basics/>

A beginner-friendly article that explains superposition, multi-qubit systems, and entanglement with practical Qiskit examples.

2. Video - Quantum Entanglement Explained

<https://www.youtube.com/watch?v=zdZjWq8R5gs>

Provides an intuitive explanation of entanglement and addresses common misconceptions.

<https://www.youtube.com/watch?v=8UrDRqVzp8w>

A visual walkthrough of how entanglement is created using quantum circuits and why it is essential in quantum information science.

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